

Computer Programming: Lecture 5

Outline

- Iteration

Iteration

- Iteration is a fundamental concept in computer programming that involves repeating a set of instructions multiple times until a certain condition is met.

Loops

- A loop repeats a sequence of statements

We have two types of loops in python:

- while loop: are used to iterate over a block of code as long as a certain condition is met.
- for loop: are used to iterate over a sequence of values. This could be a list, tuple, string, or any other iterable object.

The while loop

```
while(condition):  
    <statements>
```

The while loop

More formally, here is the flow of execution for a while statement:

- Evaluate the condition, yielding True or False.
- If the condition is false, exit the while statement and continue execution at the next statement.
- If the condition is true, execute the body

Example

Let us count 1 up to 10:

```
i = 1
while(i <= 10):
    print(i)
    i = i + 1
```

Demo: Let us write a function that can count up to a positive integer passed as an argument

Example

Let us write an accumulator loop:

Demo: Let us add a number 1 to 100

Example

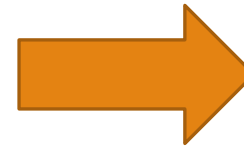
Demo: Let us write a factorial function

$$5! = 5 * 4 * 3 * 2 * 1$$

Break statement

The break statement is used to exit a loop early. It can be used in both for loops and while loops.

```
i = 0
while(i <= 10):
    if(i == 5):
        break
    print(i)
    i = i + 1
print("The current value of i is:",i)
```



```
0
1
2
3
4
The current value of i is: 5
```

Example

Guess the output of this code:

```
i = 11
```

```
while(i <= 20):
```

```
    if(i % 5 == 0):
```

```
        break
```

```
    print(i)
```

```
    i = i + 1
```

```
print("The current value of i is:",i)
```

11

12

13

14

The current value of i is: 15

Demo

Let us find the LCM of two numbers:

LCM(3,4) - 12

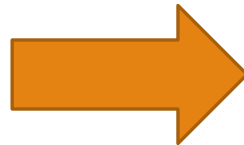
LCM(6,8) - 24

LCM(4,8) - 8

Continue statement

The continue statement is used to skip the current iteration of a loop and continue with the next iteration.

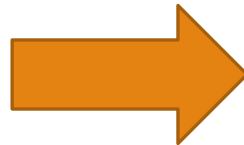
```
i = 0
while(i < 5):
    i += 1
    if(i == 3):
        continue
    print(i)
```



```
1
2
4
```

Continue Example

```
i = 0
while(i < 10):
    i = i + 1
    if(i % 2 == 0):
        continue
    print(i)
```



1
3
5
7
9

Continue statement

Guess the output of this code:

```
i = 0
```

```
while(i < 10):
```

```
    print("Hello")
```

```
    if(i % 2 == 0):
```

```
        i = i + 1
```

```
        continue
```

```
    print(i)
```

```
    i = i + 1
```

Common Errors

```
x = 10  
while(x > 0):  
    x += 1
```


Common Errors

Off by one error: A logic error that occurs in programming when an iterative loop iterates one time too many or too few.

Arises when programmer makes mistakes such as

- Fails to take account that a sequence starts at zero rather than one
- Using “less than or equal to” in place of “less than” ...

Exercises

1. Your program adds the numbers entered by a user and prints the sum of the total numbers:
 - a. Skips if the user mistakenly added a negative number
 - b. Stops adding and shows the sum if the user enters 0

2. Let's do a guessing game.
 - a. If the user a number less than 1 the program stops and thanks the user for playing
 - b. If the user guessed correctly, the program tells the user the guess is correct and stops
 - c. If the user guessed a higher number above the correct number, the program tells the user to guess lower
 - d. If the user guessed a lower number below the correct number, the program tells to the user guess higher

1. Let's do a guessing game.

- a. If the user a number less than 1 the program stops and thanks the user for playing
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The range function

The range() function in Python is used to create a sequence of numbers. The syntax for the range() function is as follows:

`range(start, stop, step)`

Range creates a sequence of number from the start to stop -1 (+1 if it is in reverse) apart by the step value

The range function

`range(0,10)` -> creates a sequence starting 0 and ending at 9

`range(0,10,2)` -> creates a sequence containing 0,2,4,6, and 8

`range(10)` -> creates a sequence starting at 0 and ending at 9

`range(10,1,-1)` -> creates a sequence starting at 10 and ending at 2

For loop

The loop variable picks up the next value in a sequence on each pass through the loop

```
for variable in sequence:  
    block of code
```

For loop example

```
for i in range(5):  
    print(i)
```



```
0  
1  
2  
3  
4
```

Example

Let us write an accumulator loop:

Demo: Let us add a number 1 to 100

Demo: Let us write a factorial function

Break and Continue

```
for letter in "python":  
    if(letter == "h"):  
        break  
    print(letter)
```

```
for letter in "python":  
    if(letter == "h"):  
        continue  
    print(letter)
```

What is the output of this code?

```
rows = 5
for i in range(1, rows + 1):
    print('*' * i)
```

```
rows = 5
for i in range(rows, 0, -1):
    print('*' * i)
```

What is the output of this code?

```
rows = 5  
for i in range(1, rows + 1):  
    print(' ' * (rows - i) + '*' * i)
```

Nested Loops

- A nested loop is a loop that is contained within another loop.
- The inner or outer loop can be any type, such as a while loop or for loop. For example, the outer for loop can contain a while loop and vice versa.
- Each time the outer loop is repeated, the inner loops are re-entered and started anew.

```
# outer loop
for element in sequence:
    # inner loop
    for element in sequence:
        # body of inner loop
    # body of outer loop
```

```
# outer loop
while condition:
    # inner loop
    while condition:
        # body of inner loop
    # body of outer loop
```

Example

What is the output of the following snippets

```
for i in range(5):  
    for j in range(2):  
        print(i,j)
```

```
i = 0  
while(i < 5):  
    j = 0  
    while(j < 2):  
        print(i,j)  
        j+=1  
    i += 1
```

What is the output of this code?

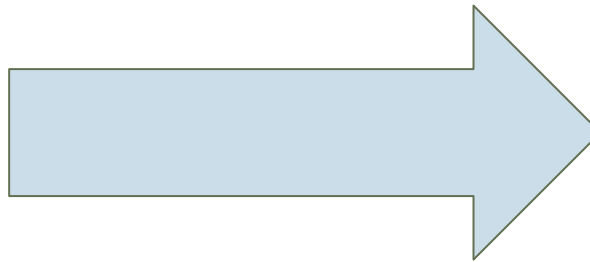
```
rows = 5
for i in range(1, rows + 1):
    for j in range(i):
        print('*', end=' ')
    print()
```

What is the output of this code?

```
rows = 5
# outer loop
while rows > 0:
    # inner loop
    cols = 3
    while cols > 0:
        # print star
        print("*", end=" ")
        # decrement cols by one
        cols -= 1
    # print new line
    print()
    # decrement rows by one
    rows -= 1
```

Example

```
rows = 5
# outer loop
while rows > 0:
    # inner loop
    cols = 3
    while cols > 0:
        # print star
        print("*", end=" ")
        # decrement cols by one
        cols = cols - 1
    # print new line
    print()
    # decrement rows by one
    rows = rows - 1
```



```
* * *
* * *
* * *
* * *
* * *
```

Question: Draw this pattern without using nested loops.

Some Loop Questions

Write a python program that accepts a positive number and prints the sum of the digits.

Write a python program that counts the frequency of a digit in a given number.

Eg. `digit_counter(1223,2)` -> 2